

An Ensemble Kalman Smoother for Online Paleoclimate Data Assimilation

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1 Introduction

1. Paleoclimate data assimilation (PDA) combines **proxies** and **climate models** to produce an optimal estimate of past climate.
2. PDA is a “**reconstruction**” problem: besides information from the past and present, information from the future is also available
3. Traditional PDA is based on a filter, which utilizes proxy information at the present time or from the **past** to the **present** time^[1].
4. This study proposes a smoother framework, which incorporates proxy information **from the past to the future**^[2].

2 Data assimilation methods

1. Integrate hybrid ensemble Kalman filter (IHEnKF): assimilate past and current proxy
2. Ensemble Kalman smoother (EnKS): based on IHEnKF, use $\mathbf{P}_{t|t+l}^{af}$ to assimilate future proxy
3. EnKS_GGF: EnKS + Localization: $\rho_{t|t+l} \circ \mathbf{P}_{t|t+l}^{af}$
4. IHEnKS: EnKS + Hybridization: $\mathbf{P}_{hyb,t|t+l}^{af} = \gamma \mathbf{P}_{t|t+l}^{af} + (1 - \gamma) \mathbf{P}_{climo,t|t+l}^f$
5. IHEnKS_GGF: EnKS + Localization + Hybridization: $\rho_{t|t+l} \circ \mathbf{P}_{hyb,t|t+l}^{af}$

3 Experimental design

1. Assimilate temperature sensitive records from PAGES2k
2. Train AI surrogate model using CESM Last Millennium Ensemble
3. Evaluate using coefficient of efficiency (CE)

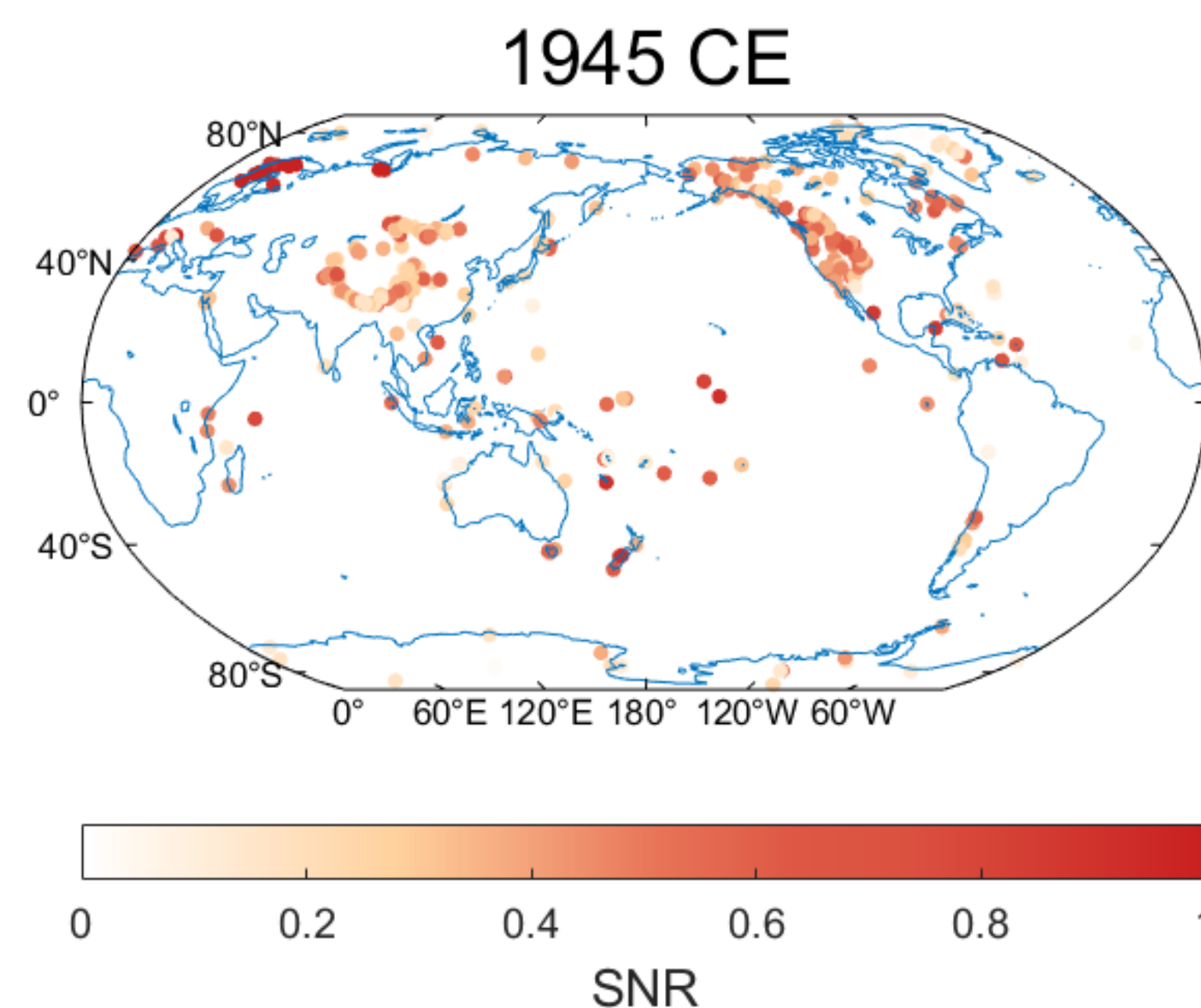


Figure 1. Spatial distributions of proxy records, colors indicate the signal-to-noise ratio (SNR).

4 Schematic overview

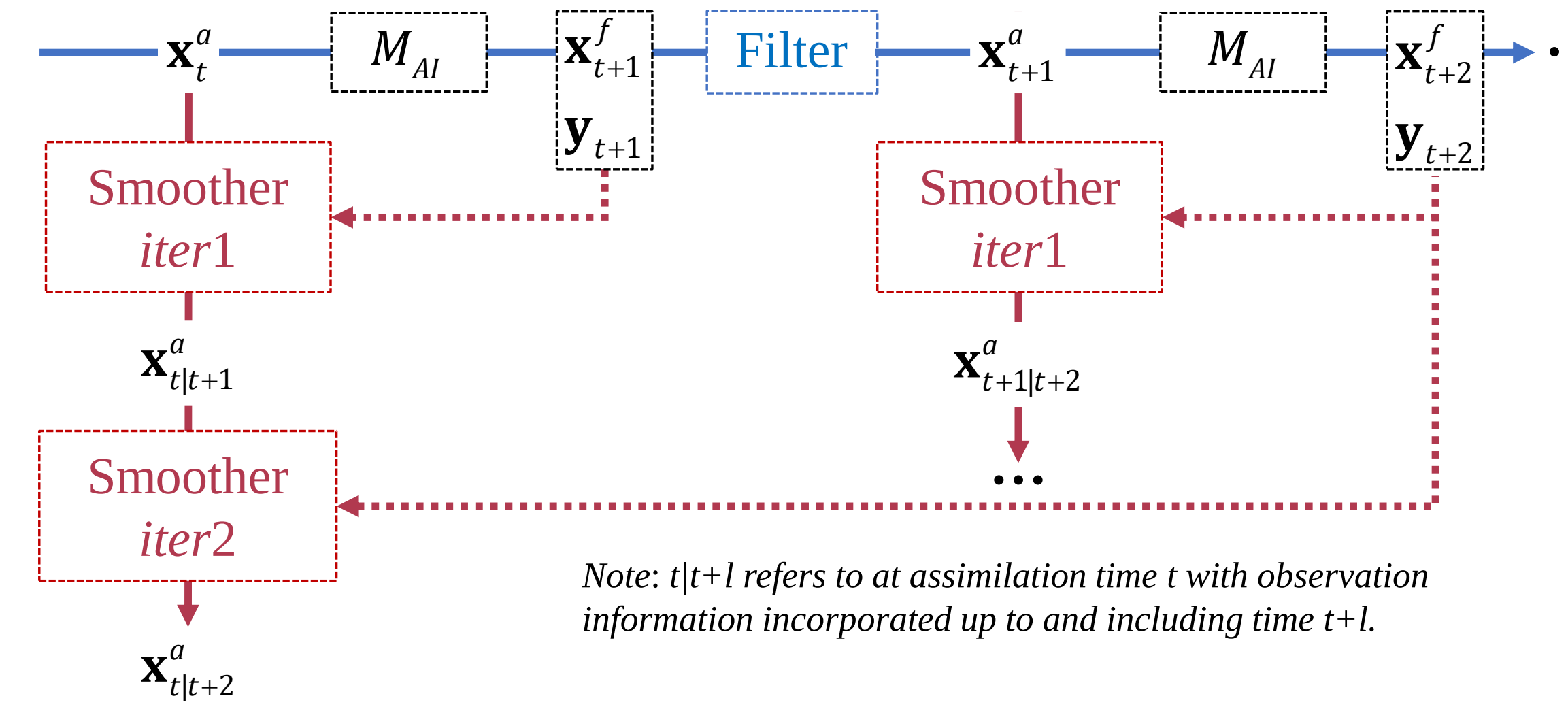


Figure 2. Schematic workflows of the filter and smoother.

5 Results: pseudoproxy experiments

1. IHEnKF has decreased reconstruction skill for the last 4 months, due to degraded forecast skill of AI model
2. EnKS improves last 4 months and degrades first 8 months, as first 8 months are far from future proxy
3. EnKS_GGF improves all 12 months, as **first 8 months are less impacted by future proxy with stronger localization**
4. IHEnKS further improves last 4 months, **as more accurate correlations are estimated by climatology**
5. IHEnKS_GGF gives the **best performance** for 12 all months

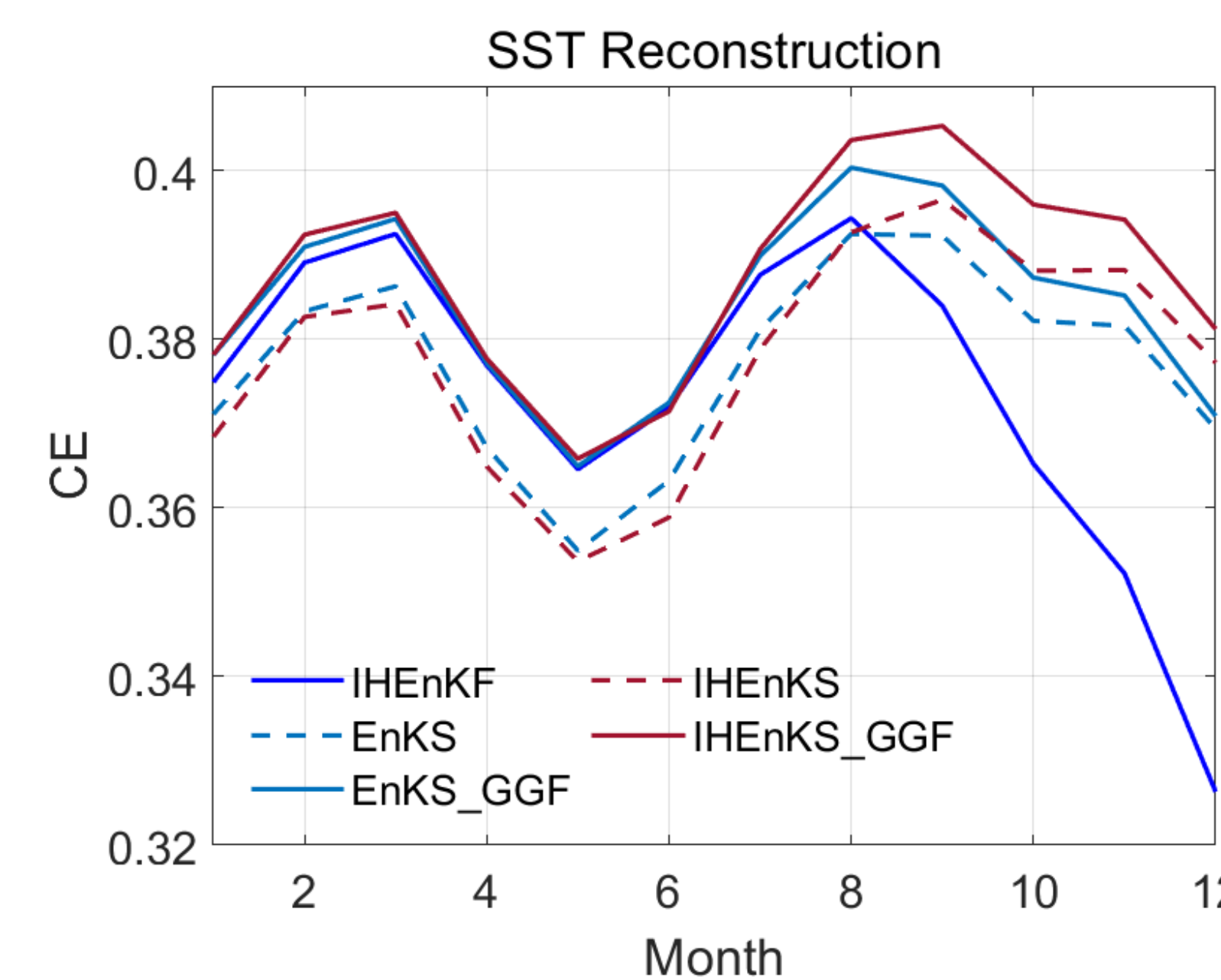


Figure 3. Area-weighted global averaged and temporally averaged CEs of monthly SST and for IHEnKF, EnKS, EnKS_GGF, IHEnKS, and IHEnKS_GGF.

Results: real proxy experiments

1. El Niño is shown in the observational product (Extended Reconstructed SST, ERSST)
2. IHEnKF does not reconstruct the El Niño, only presents a week warming in Pacific Ocean
3. IHEnKS_GGF incorporates future proxy with encoded El Niño signals, leads to a more pronounced El Niño

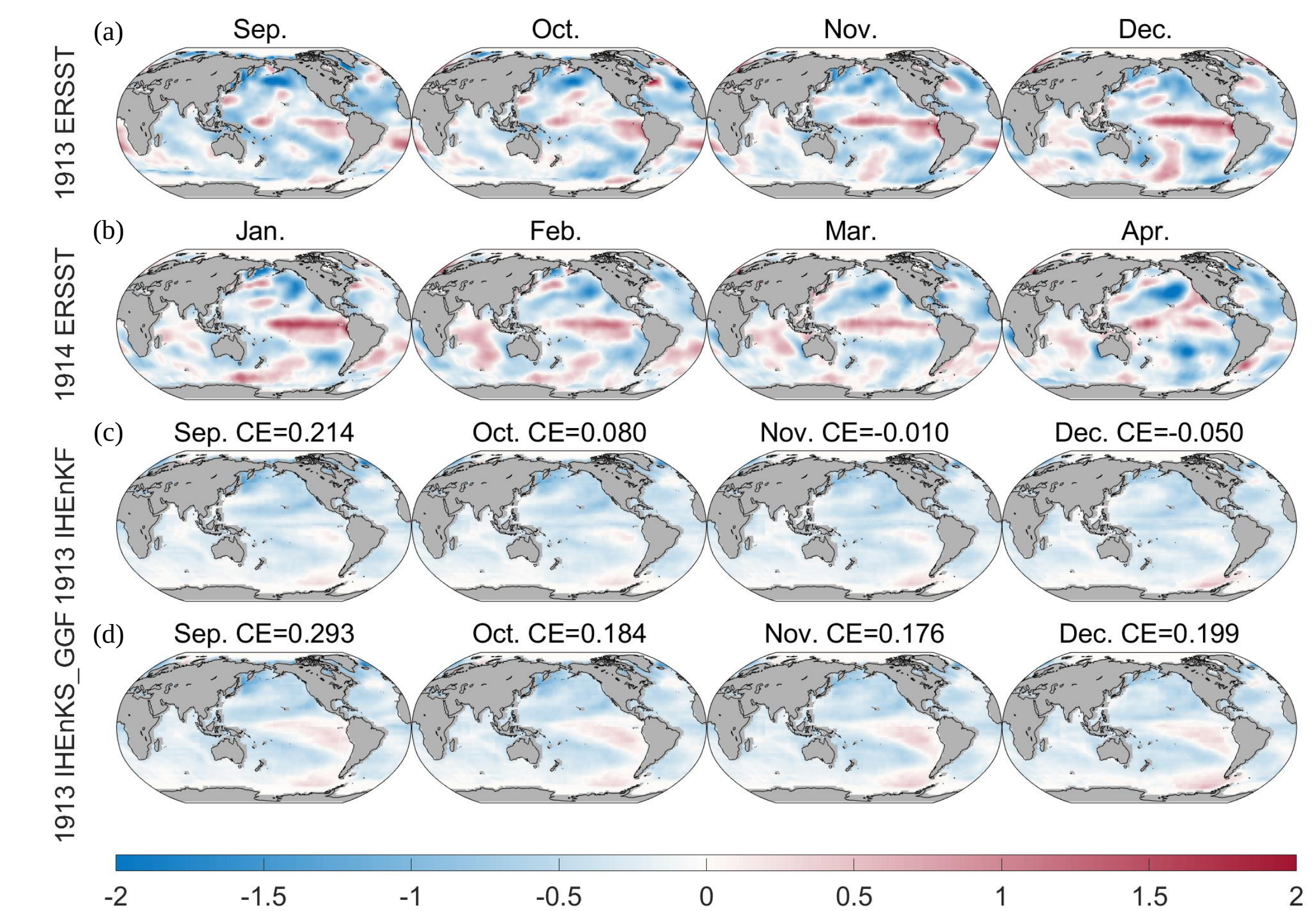


Figure 4. SST reconstruction of 1913 CE for (a) ERSST, (c) IHEnKF, and (d) IHEnKS_GGF from September to December 1913, and (b) ERSST from January to April 1914.

6 References

- [1] Sun, H., L. Lei, Z. Liu, L. Ning, and Z. Tan, 2025: An Online Paleoclimate Data Assimilation With a Deep Learning-Based Network. *J. Adv. Model. Earth Syst.*, 17, e2024MS004675, <https://doi.org/10.1029/2024MS004675>.
- [2] Sun, H., L. Lei, Z. Tan, L. Ning, and Z. Liu, 2026: An Ensemble Kalman Smoother for Online Paleoclimate Data Assimilation. *J. Adv. Model. Earth Syst.* (Under review)